

7.2 Transverse Waves: EM Spectrum & Polarisation

Question Paper

Course	CIE A Level Physics
Section	7. Waves
Topic	7.2 Transverse Waves: EM Spectrum & Polarisation
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

Question 1

A beam of unpolarised light is incident on the first of two parallel polarisers. The transmission axis of the two polarisers are initially parallel.

The first polariser is now rotated about the direction of the incident beam by an angle smaller than 90° .

Which row in the table describes the changes to the intensity and polarisation of the transmitted light?

	Intensity	Polarisation
A	no change	no change
B	no change	different
C	different	different
D	different	no change

[1 mark]

Question 2

Unpolarised light is incident on the surface of a transparent medium. The reflected light is completely plane polarised.

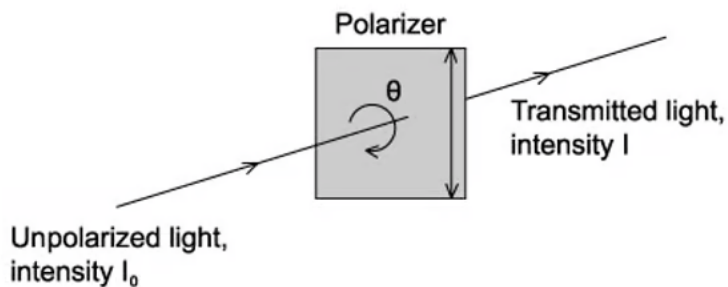
Which option correctly describes the refracted light?

- A. partially plane polarised
- B. completely plane polarised parallel to the reflected light
- C. unpolarised
- D. completely plane polarised at right angles to the reflected light

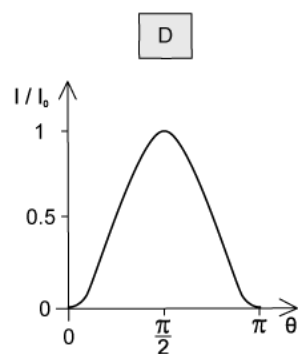
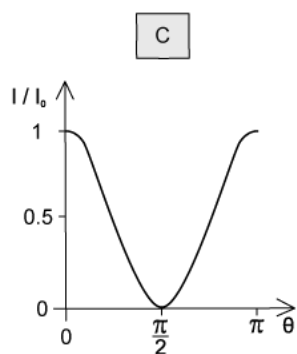
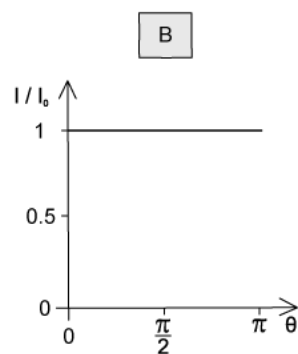
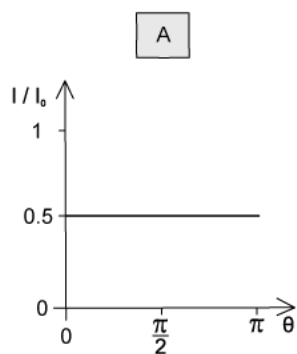
[1 mark]

Question 3

Unpolarised light of intensity I_0 is incident on a polariser that has a vertical transmission axis, as shown in the diagram below.



Which graph shows the intensity of the transmitted light ?



[1 mark]

Question 4

Plane-polarised light of amplitude a is passed through a polarising filter. The amplitude of the light emerging is $a \cos \theta$.

The intensity of the incident ray is I

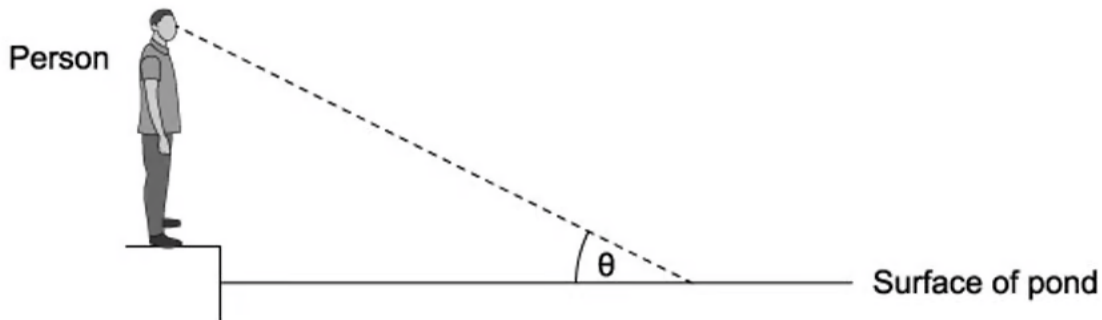
What is the intensity of the emerging light when θ is 60° ?

- A. $0.866 I$
- B. $0.750 I$
- C. $0.500 I$
- D. $0.250 I$

[1 mark]

Question 5

A person wearing polarising sunglasses stands at the edge of a pond in bright sunlight.



The surface of the pond is flat, and the person has a line of sight to the surface of angle θ . The refractive index of the pond water is n .

What is the value of θ for which the light would be a minimum when it reaches the person's eye?

- A. $\tan^{-1}\left(\frac{1}{n}\right)$
- B. $\cos^{-1}\left(\frac{1}{n}\right)$
- C. $\tan^{-1}(n)$
- D. $\cos^{-1}(n)$

[1 mark]

Question 6

Unpolarised light is incident on a polariser. The light transmitted by the first polariser is then incident on a second polariser. The polarising axis of the second sheet is 60° to that of the first.

The intensity emerging from the second polariser is I_2 .

Which of the following gives the intensity incident on the first polariser?

A. $\frac{I_2}{4}$

B. $\frac{I_2}{8}$

C. $8I_2$

D. $4I_2$

[1 mark]

Question 7

A vertical aerial transmits radio waves that have been plane-polarised. The receiving aerial is tilted from the vertical at an angle θ . The emitted wave is in a plane perpendicular to the direction of arrival and arrive with an amplitude A .

Which of the following expressions is proportional to the power delivered by the aerial?

A. $A \cos \theta$

B. $A^2 \cos^2 \theta$

C. *Zero*

D. $A^2 \sin^2 \theta$

[1 mark]

Question 8

Unpolarised light of intensity I_0 is incident on the first of two polarising sheets. Initially the planes of polarisation of the sheets are perpendicular.

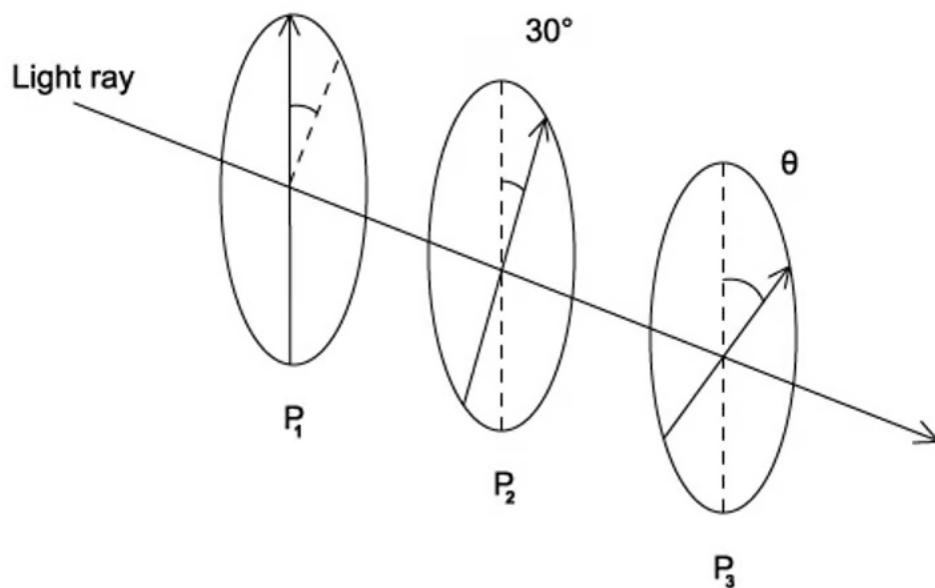
Which sheet must be rotated and by what angle θ , so that light intensity $\frac{I_0}{4}$ can emerge from the second sheet?

	Rotated sheet	Angle θ of rotation
A	1 or 2	$\cos^{-1} \frac{1}{2}$
B	1 or 2	$\cos^{-1} \frac{\sqrt{2}}{2}$
C	2 only	$\cos^{-1} \frac{1}{2}$
D	1 only	$\cos^{-1} \frac{\sqrt{2}}{2}$

[1 mark]

Question 9

The diagram below shows a ray of unpolarised light with an intensity of 72 W m^{-2} passing through three polarising films, P_1 , P_2 and P_3 .



The polarising axis of each polarising film is shown by an arrow. The polarising film P_1 and P_2 are fixed, with their axis of polarisation at 30° to one each other.

If the light emerging from the third filter has an intensity of 13.5 W m^{-2} what is angle θ ?

- A. 60°
- B. 90°
- C. 30°
- D. 45°

[1 mark]

Question 10

The angle between a polariser and analyser is 30° . The intensity of light transmitted by the analyser is greater than the intensity transmitted by the polariser.

How much smaller is the intensity transmitted?

A. $\frac{1}{2}$

B. $\frac{3}{4}$

C. $\frac{1}{4}$

D. 0

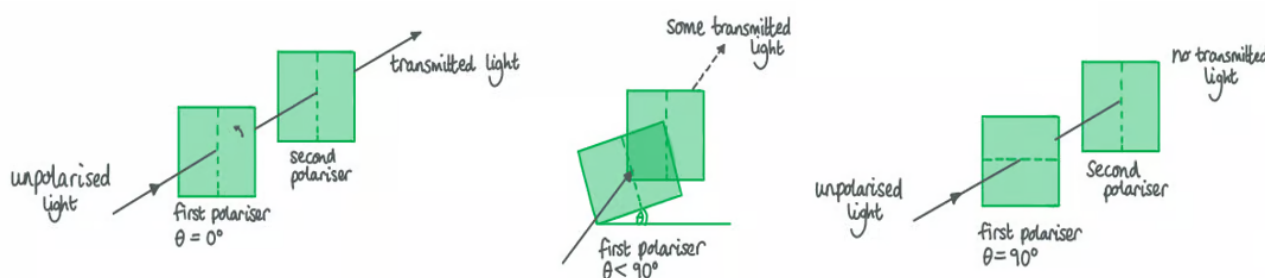
[1 mark]

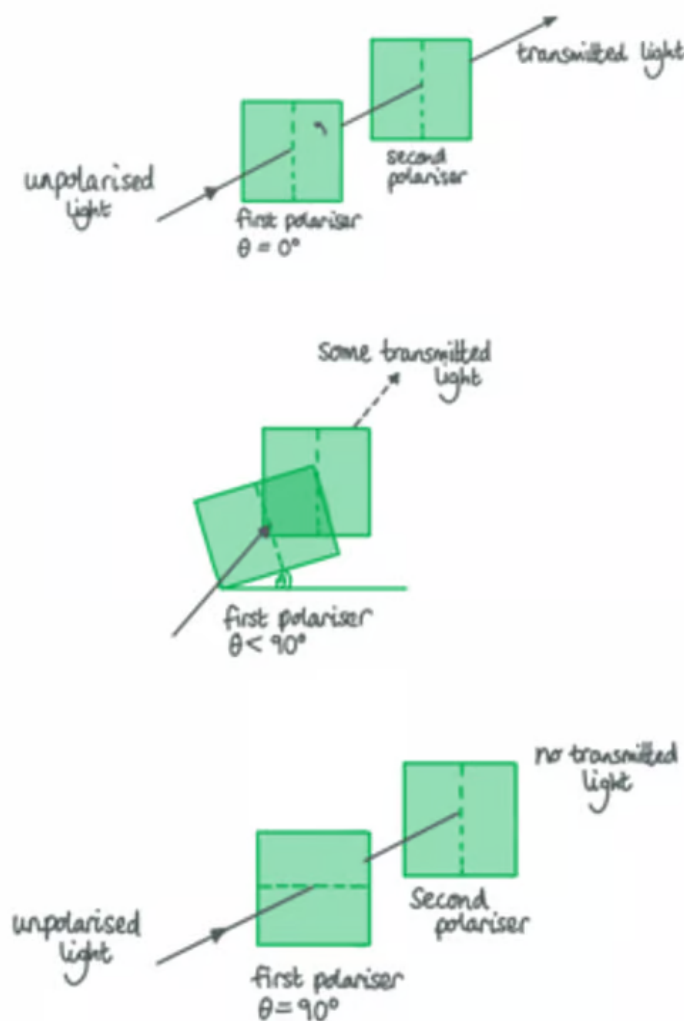
Model Answers: Hard

1

The correct answer is **D** because:

- Polarisation is the process by which the oscillations are made to occur in one plane only
- So, the first polariser causes the polarisation of the light to be oriented by angle θ
- The first polariser sets the polarisation of the wave and the second only lets through light of that polarisation
- If the second polariser was aligned at 90° to the first, it would block the light completely
- However, since the second polariser is aligned at an angle smaller than 90° to the first, it will allow some light of polarisation with angle θ through and hence reduce the intensity of the light
- It is not possible to polarise already polarised light, so the polarisation does not change

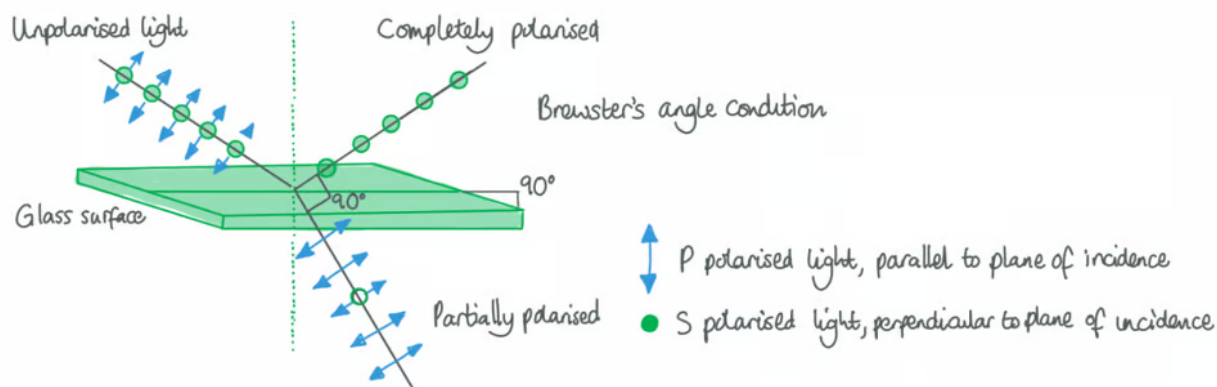




2

The correct answer is **A** because:

- At the Brewsters angle, the reflected light is plane polarised
 - Brewster's angle is an angle of incidence at which light with a particular polarisation is perfectly transmitted through a surface
- The refracted light might not be polarised, there will be some light in the same plane as the reflected light, but less of it
- The transmitted light is therefore partially plane polarised



B & D are incorrect because the light is not completely plane polarised

C is incorrect because some of the light will be polarised

3

The correct answer is **A** because:

- Polarisation is the process by which the oscillations are made to occur in one plane only
- As the light passes through the first polariser half of the intensity $\left(\frac{I_0}{2}\right)$ of the ray is lost
 - This is known as the half rule

The other graphs do not recognise that the incoming light is unpolarised, so the other graphs are incorrect

4

The correct answer is **D** because:

- Polarisation is the process by which the oscillations are made to occur in one plane only
- The light entering the filter is plane polarised
- When the filter is moved 60° to the vertical the intensity of the light will be

$$\circ I = \cos^2 60 = 0.250$$

- Therefore, the intensity of the emerging light is 0.250 I

A is incorrect because the intensity of the emerging ray if unpolarised light is incident on a filter with an angle of 30° is 0.866 I

B is incorrect because the intensity of the emerging ray if polarised light is incident on a filter with an angle of 30° is 0.750 I

C is incorrect because the intensity of the emerging ray unpolarised light is incident on a filter with an angle of 60° is 0.500 I

5

The correct answer is **A** because:

- Polarisation is the process by which the oscillations are made to occur in one plane only
- Brewster's angle is an angle of incidence at which light with a particular polarisation is perfectly transmitted through a surface
- Brewster's angle is given by

$$\circ \tan \theta_B = \frac{n_2}{n_1}$$

- where n_1 is the refractive index of the initial material and n_2 is the refractive index of the material scattering the light

- Rearranging this gives $\theta_B = \tan^{-1} \left(\frac{n_2}{n_1} \right)$

- We are told the refractive index of the pond water is $n_1 = n$
- When the light leaves the water's surface, it travels through the air which has a refractive index of $n_2 = 1$

$$\circ \text{So, } \theta_B = \tan^{-1} \left(\frac{1}{n} \right)$$

B & D are incorrect because these use the incorrect trigonometry function of cos

C is incorrect because θ is the angle to the surface not the angle of incidence

6

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6

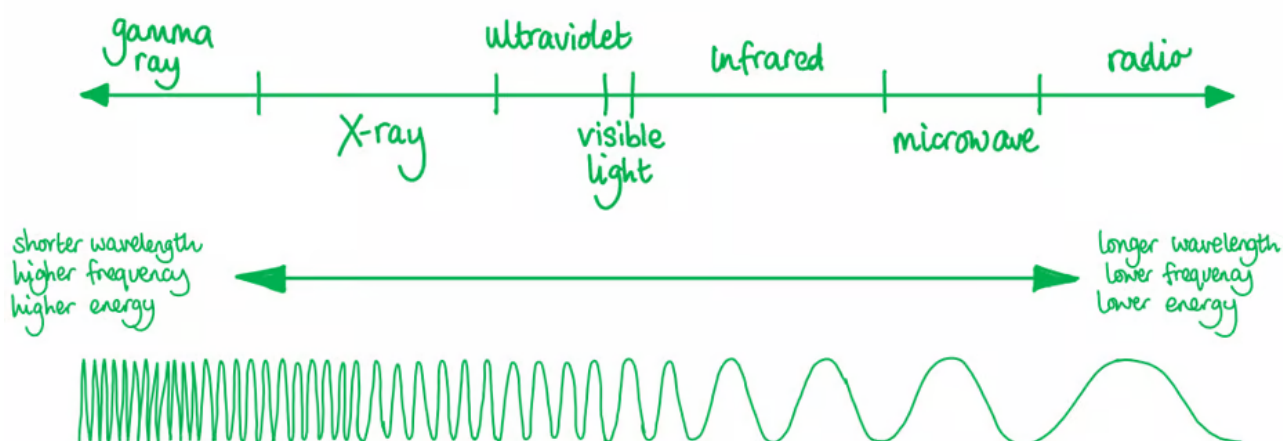
The correct answer is **C** because:

- Polarisation is the process by which the oscillations are made to occur in one plane only
- As the light passes through the first polariser half of the intensity $\left(\frac{I_0}{2} \right)$ of the ray is lost
 - This is known as the half rule
- Malus's law allows us to calculate the intensity of a beam of plane polarised light, it is given by
 - $I = I_0 \cos^2 \theta$
- The intensity of light on the second polariser is equal to $= I_0 \cos^2 60 = \frac{I_0}{4}$
- Combining the intensity drops from both polarisers, the total intensity of the light after it leaves the second polariser is $\frac{1}{2} \times \frac{I_0}{4} = \frac{I_0}{8}$
 - Therefore, the intensity incident on the first polariser is $8I_0$

7

The correct answer is **B** because:

- The radio waves will be oscillating in the direction of the antenna that emitted it, half of this oscillation is equal to the amplitude A of the radio wave
- Intensity is proportional to the square of the amplitude
 - $I \propto A^2$
- The receiving aerial is at an angle θ to the vertical
- Malus law states that the transmitted intensity of the plane polarised light passing through a polariser is directly proportional to the square of the cosine of the angle through which the polariser is rotated from the position which gives the maximum intensity
 - $I \propto \cos^2 \theta$
- Since intensity is the power transferred per unit area, intensity is directly proportional to power
 - Therefore, $I \propto P \propto A^2 \cos^2 \theta$



Transmitting Aerial and Plane of Polarisation



A & D are incorrect because Malus' law tells us intensity $\propto \cos^2 \theta$

C is incorrect because only if zero power were delivered would there be no signal

8

The correct answer is **B** because:

- Polarisation is the process by which the oscillations are made to occur in one plane only
- As the light passes through the first sheet half of the intensity of the ray is lost, this is known as the half rule
- The second filter will need to reduce the intensity by another half in order for the emerging light to be $\frac{I_0}{4}$
- An angle of 45° will reduce the intensity of the light by half $\left(\frac{I_0}{2}\right)$ so the angle of rotation must be 45° and this gives $\cos 45 = \frac{\sqrt{2}}{2}$
- Either filter could be moved as both will reduce the intensity by half each time the light passes through

A is incorrect because $\cos^{-1} \frac{1}{2}$ would give an angle of 60° . This would reduce the intensity of the light greater than 50%

C & D are incorrect because either filter could be rotated

9

The correct answer is **D** because:

- Polarisation is the process by which the oscillations are made to occur in one plane only
- As the light passes through the first sheet half of the intensity $\left(\frac{I_0}{2}\right)$ of the ray is lost, this is known as the half rule
- Malus law states that the transmitted intensity of the plane polarised light passing through a rotatable polariser is directly proportional to the square of the cosine of the angle through which the polariser is rotated from the position which gives the maximum intensity.
 - $I \propto \cos^2 \theta$
- Intensity of light from $P_1 = \frac{I_0}{2} = \frac{72}{2} = 36 \text{ W m}^{-2}$
- Intensity of light from $P_2 = 36 \cos^2 30 = 27 \text{ W m}^{-2}$
- Intensity of light from $P_3 = 27 \cos^2 45 = 13.5 \text{ W m}^{-2}$
 - Therefore, the angle of the third polariser is equal to $\theta = 45^\circ$

10

The correct answer is **B** because:

- Malus law states that the transmitted intensity of the plane polarised light passing through a rotatable polariser is directly proportional to the square of the cosine of the angle through which the polariser is rotated from the position which gives the maximum intensity
 - $I \propto \cos^2 \theta$
 - Where θ is the angle through which a polariser is rotated from the position which gives the maximum intensity
- When a polariser is placed at 30° the intensity is reduced by 25%, so the intensity of the light will be $\cos^2 30 = \frac{3}{4}$

A is incorrect because the intensity of the light is not reduced by 50%

C is incorrect because the light is reduced by $\frac{1}{4}$ by the 30° polariser

D is incorrect because some of the light will get through the polariser

